Metrics at a glance

Judging probabilistic PV disaggregation · C3-A7 summary, for use in C3-A8

pv_i: true PV · $\hat{pv}_i$: estimate · C: installed capacity (kWp)

POINT ESTIMATE (MEDIAN, MEAN OR KDE MODE): HOW FAR OFF?

RMSE root mean squared error

MEASURES	Typical size of the error; squaring punishes large errors more.
UNIT	kW, or kW/kWp to compare households
GOOD	Lower; 0 is perfect
TRAP	Grows with system size. Compare households per kWp, or use MASE.

MASE mean absolute scaled error

MEASURES	Error relative to a naive persistence estimate (the previous half-hour's true PV).
UNIT	None
GOOD	Below 1 beats persistence
TRAP	Persistence needs the true PV, which a DSO never has: it is a ruler, not a rival. Smooth clear days make it hard to beat.

MAPE / SMAPE (symmetric) percentage error

MEASURES	Relative error. SMAPE divides by true + estimate, treating over- and underestimation alike.
UNIT	%
GOOD	Lower
TRAP	Explodes near zero (dawn, dusk, dull days). Compute only around midday (10:00–15:00), as in [1].

PREDICTION INTERVALS: HOW HONEST AND HOW USEFUL IS THE UNCERTAINTY?

PICP PI coverage probability

MEASURES	Calibration: share of true values inside the PI.
UNIT	%
GOOD	Close to the promised level (≈ 80% for an 80% PI)
TRAP	Higher is not better: very wide intervals reach 100%. Always read it with PINAW.

PINAW PI normalised average width

MEASURES	Sharpness: mean width of the PI divided by C.
UNIT	kW/kWp
GOOD	Lower, but only if PICP is close to its target
TRAP	A narrow but overconfident interval looks sharp and is wrong.

Pinball loss quantile loss

MEASURES	Each quantile, with an asymmetric penalty; averaged over q = 0.1 … 0.9 it rewards calibration and sharpness together.
UNIT	kW, or kW/kWp to compare households
GOOD	Lower
TRAP	Weights q and 1 − q are easy to swap: q applies when the truth lies *above* the quantile.

Calibration diagram
Observed vs nominal coverage at 10 … 90%. On the diagonal: calibrated. Below: overconfident (too narrow). Above: underconfident (too wide).

Daylight only
Compute PICP, PINAW and pinball loss where irradiance > 0. Night zeros are covered by every interval and make any model look calibrated.

Many days, many households
One day can land anywhere by chance: over 300 days, a single day's 80% PI covered 33–100% of the truth.

[1] M. Jené-Vinuesa et al., "Conditional diffusion modeling for probabilistic behind-the-meter PV disaggregation," *Energy and AI*, 2026, doi: 10.1016/j.egyai.2026.100692, Section 3.2. AI4GreenDeal · C3 Conditional diffusion modeling for energy disaggregation.